ANEX JOHNSON

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SUMMARY

Versatile Product Designer with expertise in 2D/3D drafting, isometric drawing, parametric and surface modeling, sheet metal design, assembly design, and simulation-based analysis. Experienced in performance analysis and design optimization through detailed modeling and simulation techniques.

SKILLS

- **AutoCAD 2D**: Proficient in AutoCAD 2D drafting and isometric mode for mechanical and architectural designs. Skilled in Xrefs, BOMs, layouts, viewports, blocks, and annotation tools for clear and accurate documentation.
- **AutoCAD 3D:** Developing 3D models of components and drafting. Experienced with visual styles, UCS management,
- **CATIA:** Proficient in CATIA for Part design, Assembly modelling, Wireframe and Surface modelling, Sheet metal design, Sketch tracer, Drafting, and Basic animation, with hands-on experience in creating detailed 3D engineering models.
- **ANSYS:** Skilled in ANSYS for structural, thermal, thermal stress, modal, buckling, and fatigue analysis, with hands-on experience in **DesignModeler** and **SpaceClaim** for geometry creation, editing, defeaturing, repair, and simplification, along with meshing, boundary conditions, solver setup, and post-processing.
- **SOLIDWORKS:** Skilled in SolidWorks for 3D modelling, assemblies, drafting, sheet metal, surface modelling, motion studies, and high-quality rendering for presentations.
- **CypCut**: Proficient in CypCut for CNC Fiber laser cutting, CAD file import, kerf compensation, nesting, toolpath generation, parameter setting, lead-in/lead-out editing, sheet alignment, cutting optimization, layer management, machine control.
- **FUSION 360**: Developing 3D models, assemblies, simulation, CAM, and rendering.
- **GD&T Knowledge, Material Knowledge, Process Optimization, Inventory Management, Quality Control, Metrology and Measurement.**
- **Soft Skills**: Communication, Teamwork, Problem-solving, Adaptability, Quick Learning, Critical thinking, Time management.

EXPERIANCE

– **2D & 3D CAD Designer / Draftsman – NCK METALKO LLP Kochi, Kerala,** *August 2025 – October 2026*
  - Designed 3D models**, Sheet metal** components, and 2D production drawings for **CNC Fiber laser cutting** and fabrication using SOLIDWORKS and AutoCAD.
  - Utilized CypCut for nesting optimization, kerf compensation, and toolpath generation to reduce material waste and improve cutting efficiency.
  - Prepared cost estimations using ERP software for production and fabrication requirements.

EDUCATION

- Bachelor's degree in mechanical engineering, APJ Abdul Kalam Technological University Kerala India, *2020 - 2024*
- Expert certification in product design and analysis, CADD Centre KOCHI, *2024 - 2025*

PROJECTS

- **Sustainable Wire Insulation Removal System**
  – Developed a gear-controlled system for separating insulation from copper and aluminium conductors.
  – Created the complete 3D mechanical model in Autodesk Fusion and prepared technical workflow presentations.
- **Iron Man Helmet (CATIA)**
  – Developed a 3D Iron Man helmet model in CATIA using advanced surface modeling and Sketch Tracer for accurate complex curves and form design.
- **Motor Blower (AutoCAD 2D)**
  – Individually created each component Motor Blower using given dimensions and arranged all parts in an isometric view to accurately represent the final assembled system.
- **Industrial Trolley (CATIA and ANSYS)**
  – Modeled and assembled a trolley system in CATIA and imported it into ANSYS for meshing and simulation.
  – Performed static structural and buckling analysis under applied loads and gravity, achieving 2.46 mm maximum deformation, 128.58 MPa von-Mises stress, safety factor above 1.94, and stable buckling performance.